

APPM 6470 Project Proposal

Impact of Crowding on Diffusive Search: A Fokker–Planck and First Passage Time Approach

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Topic Area

Application of PDE methods (Fokker–Planck equations and first passage time analysis) to stochastic search processes in heterogeneous environments. The Fokker–Planck equation governs the evolution of the probability density (forward problem), while the backward equation governs the mean first passage time directly; both formulations will be needed.

Motivation

The primary motivation is foraging behaviour in crowded environments: a bee navigating within a nest encounters spatially varying traffic density, which reduces its effective diffusivity in congested regions. Similar phenomenology arises in cellular consortia and microbial colonies. In all these settings the diffusion coefficient depends on the local crowder density rather than being constant.

Crowding affects how particles explore space, often slowing transport and altering the time required to locate a target. We focus on the regime of slowly varying crowder density, giving spatially dependent but otherwise normal diffusion $\langle x^2 \rangle \sim t$ with a spatially varying coefficient $D(x)$, and defer strong or dynamic crowding (which leads to anomalous subdiffusion $\langle x^2 \rangle \sim t^\alpha$, $\alpha < 1$) to the stretch goals.

The central question is: given a partially absorbing target at $x = 0$ with discoverability κ , does a spatially dependent diffusivity $D(x)$ speed up or slow down first passage relative to the homogeneous case $D \equiv D_0$? To address this, we will analyse the mean first passage time (MFPT) and investigate how transport (through $D(x)$) interacts with detection (through κ).

A key aspect is the potential competition between these effects when κ depends on the local diffusivity at the target. For example, if $\kappa = \kappa_0/D(0)$, then lower diffusivity near the trap may both slow arrival and increase detectability, suggesting a trade-off. We aim to determine whether an optimal value of $D(0)$ exists that balances these effects.

More generally, we will compare different functional relationships between κ and $D(0)$, including κ independent of D , $\kappa \propto D(0)$, and $\kappa \propto 1/D(0)$, to understand how microscopic assumptions influence macroscopic search efficiency.

Model Overview

We derive the Fokker–Planck equation from a lattice random walk with position-dependent jump rates. In the symmetric exclusion process (SEP), volume exclusion acts at the arrival site rather

than the departure site, which leads to Fick's form

$$\frac{\partial p}{\partial t} = \frac{\partial}{\partial x} \left[D(x) \frac{\partial p}{\partial x} \right].$$

We adopt Fick's form throughout, justified by the SEP argument, and note that it has the additional practical advantage that the backward operator is self-adjoint.

Let $T(x)$ denote the mean first passage time to the trap starting from position x . The forward operator for Fick's form is $\mathcal{L}p = \partial_x[D\partial_x p]$.

The MFPT satisfies $\mathcal{L}^\dagger T = -1$. The backward equation is therefore

$$\frac{d}{dx} \left[D(x) \frac{dT}{dx} \right] = -1, \quad x \in (-\pi, \pi) \setminus \{0\},$$

with periodic boundary conditions $T(-\pi) = T(\pi)$ and $T'(-\pi) = T'(\pi)$.

Solving this equation yields an explicit expression for the MFPT, whose structure we will analyse to understand how spatial variations in $D(x)$ influence search efficiency.

Case Study

To illustrate the general theory, we consider a simple piecewise linear ('tent') profile for $D(x)$, which allows explicit evaluation while capturing key features such as enhanced or reduced diffusivity near the target. We take

$$D(x) = D_0 + D_1 \left(1 - \frac{|x|}{\pi} \right), \quad x \in [-\pi, \pi],$$

so that D peaks at the trap (value $D_0 + D_1$) and decreases linearly to D_0 at $x = \pm\pi$.

This profile provides a tractable setting in which the MFPT can be computed explicitly from the backward equation. In particular, the dependence on $D(0) = D_0 + D_1$ allows us to investigate how local diffusivity near the trap influences search efficiency and interacts with different choices of $\kappa(D(0))$. By varying the sign of D_1 , we will compare cases where diffusivity is enhanced or reduced near the target.

Goals

1. Derive the Fokker–Planck equation from a lattice random walk and justify Fick's form using the SEP exclusion argument.
2. Derive the backward equation on the ring with partial absorption, establish the flux-jump condition, and obtain an explicit expression for the mean first passage time (MFPT).
3. For the tent profile, compute the MFPT explicitly and compare with the homogeneous baseline under the constraint $\int_{-\pi}^{\pi} D(x) dx = 2\pi D_0$.
4. Verify analytical results using a finite-difference solution of the Fokker–Planck equation.

Preliminary Plan

1. **Derivation.** Lattice walk to Fokker–Planck; SEP argument for Fick's form; SDE interpretation.

2. **Backward equation.** Adjoint derivation; ring geometry; flux-jump condition; solvability argument leading to an explicit constant and expression for $T(0)$.
3. **Case studies.** Piecewise linear tent profile via rational integration.
4. **Verification with Numerics.** Finite-difference verification.

References

1. C. W. Gardiner, *Handbook of Stochastic Methods*, Springer.
2. G. A. Pavliotis, *Stochastic Processes and Applications*, Springer.
3. S. Redner, *A Guide to First-Passage Processes*, Cambridge University Press.
4. H. Risken, *The Fokker–Planck Equation*, Springer.